

Apparent pairs and optimal cycles

Ulrich Bauer

Technical University of Munich (TUM)

Decemner 12, 2025

SALTO Kick-Off Workshop *Higher-order interactions at the crossroads of geometry and topology*

Laboratoire de Mathématiques d'Orsay, Université Paris-Saclay

Funded by



Deutsche
Forschungsgemeinschaft
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SFB
TRR
109 | Discretization
in Geometry
and Dynamics



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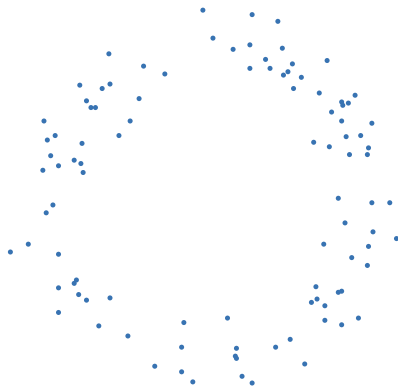


Persistence computation

Vietoris–Rips complexes

For a metric space X , the *Vietoris–Rips complex* at $t > 0$ is the simplicial complex

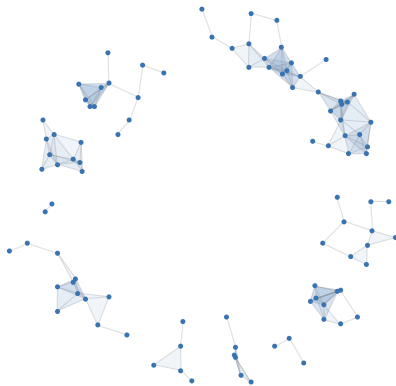
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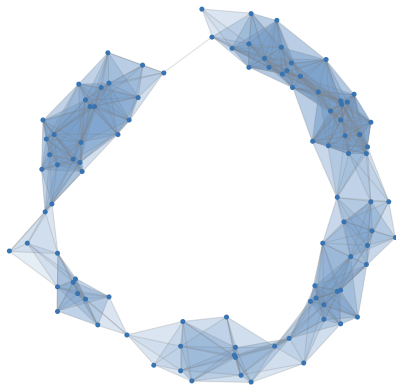
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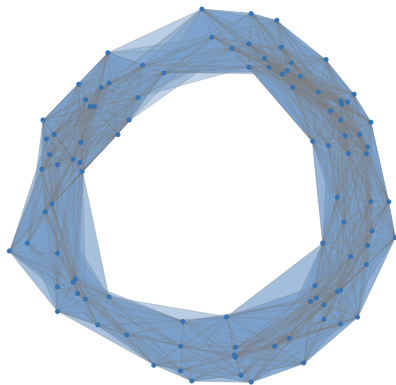
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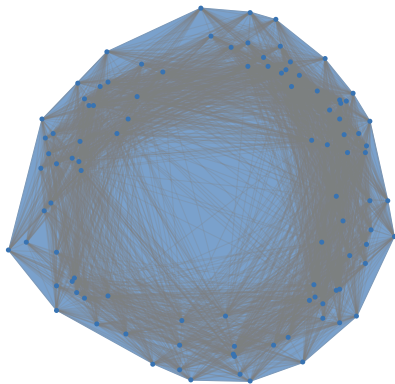
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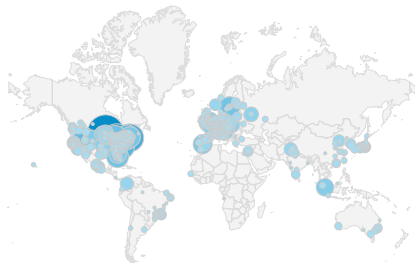
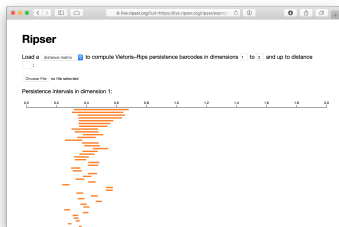
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Ripser: software for computing Vietoris–Rips persistence barcodes

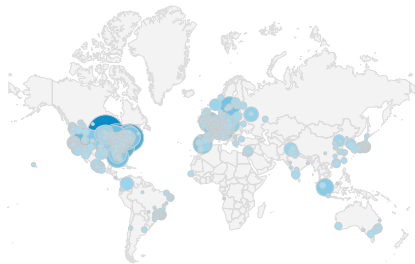
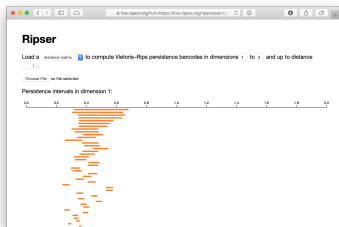
Open source software (ripser.org)



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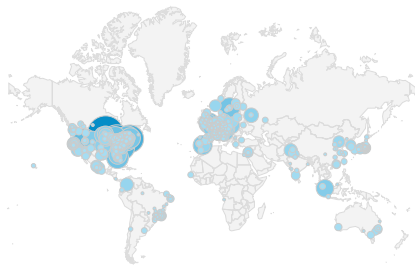
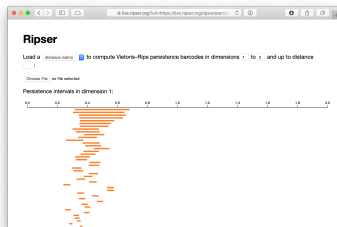
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Efficient matrix algorithm based on

- *clearing*: avoiding unnecessary column operations
- computing persistent *cohomology*

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Efficient matrix algorithm based on

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Computational improvements based on

- *implicit matrix representations*, avoiding storage of boundary matrix
- *apparent pairs*, connecting persistence to discrete Morse theory

Apparent pairs

Ripsper uses the following pairing of simplices (breaking ties in the filtration lexicographically):

Definition (B 2016, 2021)

In a simplexwise filtration $(K_i = \{\sigma_1, \dots, \sigma_i\})_i$, two simplices (σ_i, σ_j) form an *apparent pair* if

- σ_i is the latest proper face of σ_j , and
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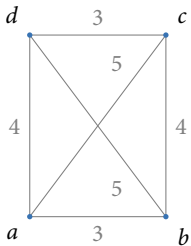
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Proposition (B 2021)

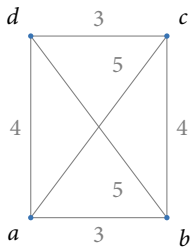
The apparent pairs are both

- *persistence pairs (creating/destroying a feature in homology) and*
- *gradient pairs (in the sense of discrete Morse theory).*

Apparent pairs of the diameter-lexicographic filtration

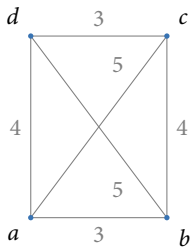


Apparent pairs of the diameter-lexicographic filtration



$$\partial_1 = \begin{matrix} & \begin{matrix} (a,b):3 \\ (c,d):3 \\ (a,d):4 \\ (b,c):4 \\ (a,c):5 \\ (b,d):5 \end{matrix} \\ \begin{pmatrix} 1 & & 1 & & 1 & \\ \mathbf{1} & & & 1 & & 1 \\ & 1 & & 1 & 1 & \\ & \mathbf{1} & 1 & & & 1 \end{pmatrix} & \begin{matrix} (a) \\ (b) \\ (c) \\ (d) \end{matrix} \end{matrix}$$

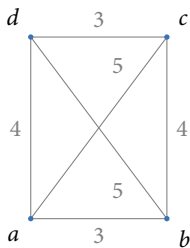
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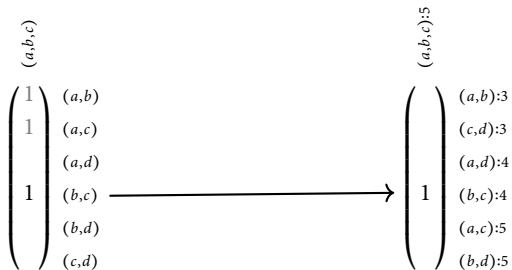
A shortcut for finding pivots

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Shortcut for finding the pivot (latest) facet of a simplex τ :

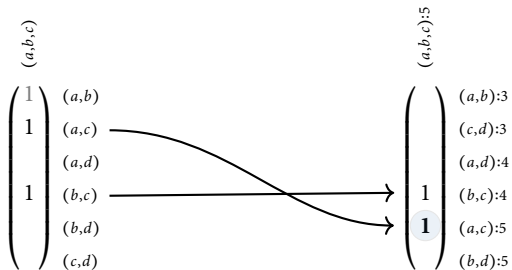
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Shortcut for finding the pivot (latest) facet of a simplex τ :

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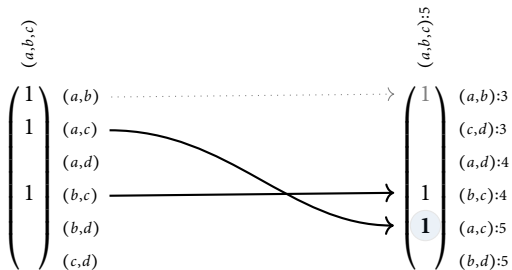
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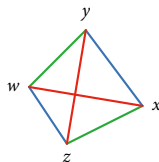
Hyperbolic collapses

Gromov-hyperbolicity

Definition (Gromov 1988)

A metric space X is δ -hyperbolic (for $\delta \geq 0$) if for all $w, x, y, z \in X$ we have

$$d(w, x) + d(y, z) \leq \max\{d(w, y) + d(x, z), d(w, z) + d(x, y)\} + 2\delta.$$



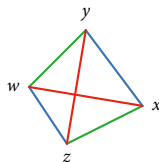
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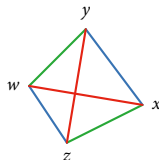
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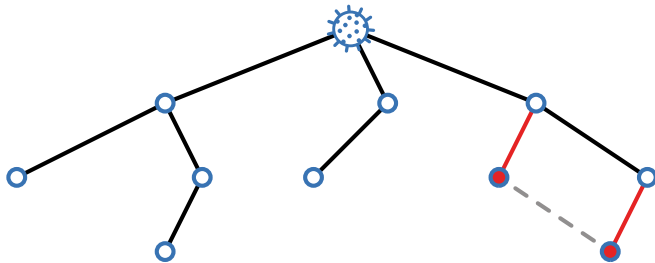
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- Metric trees and their subspaces are precisely the 0-hyperbolic spaces.

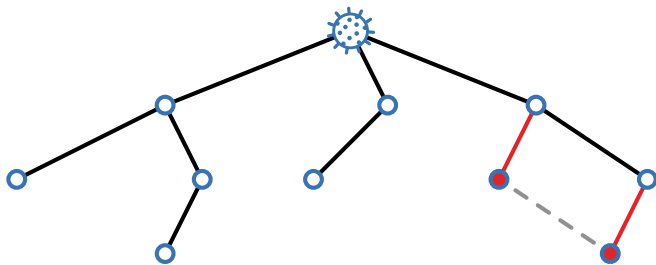


Topology of viral evolution



Joint work with: A. Ott, M. Bleher, L. Hahn (Heidelberg), R. Rabadan, J. Patiño-Galindo (Columbia), M. Carrière (INRIA)

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Observation: Ripser runs unusually fast on genetic distance data

- SARS-CoV2 RNA sequences (spike protein)
- 25556 data points (2.8×10^{12} simplices in 2-skeleton)
- 120 s computation time (with data points ordered appropriately)

Rips complexes of hyperbolic spaces

Theorem (Rips; Gromov 1988)

Let X be a δ -hyperbolic geodesic metric space. Then $\text{Rips}_t(X)$ is contractible for all $t \geq 4\delta$.

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$$\text{Rips}_t(X) \searrow \{*\}$$

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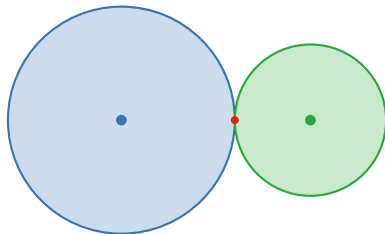
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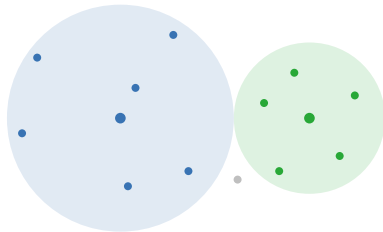
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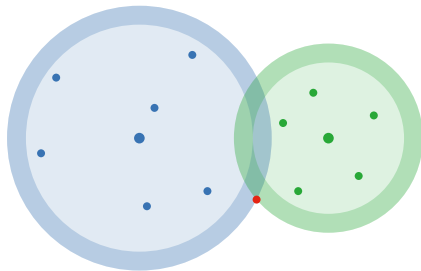
Geodesic defect



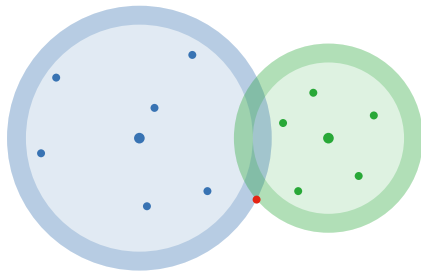
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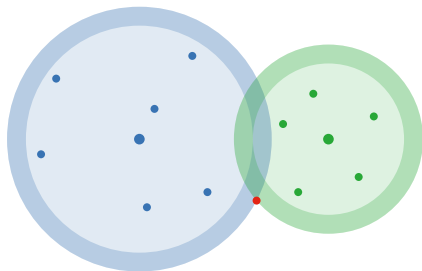
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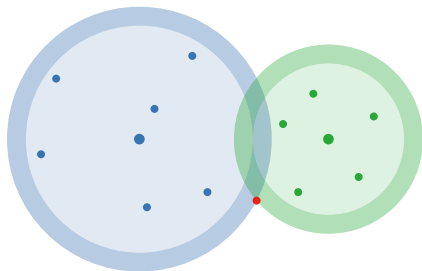


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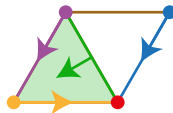
We call the infimum of all such ν the *geodesic defect* of X .

Discrete Morse theory

Definition (Forman 1998)

A *discrete vector field* on a cell complex is a partition of the set of simplices into

- singleton sets $\{\phi\}$ (*critical cells*), and
- pairs $\{\sigma, \tau\}$, where σ is a facet of τ .

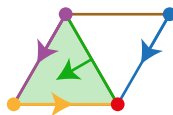


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1

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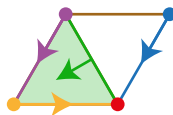
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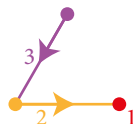
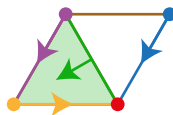
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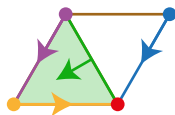
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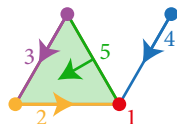
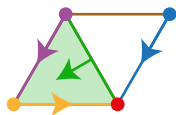
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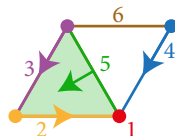
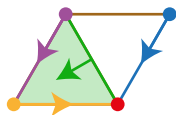
Definition (Forman 1998)

A *discrete vector field* on a cell complex is a partition of the set of simplices into

- singleton sets $\{\phi\}$ (*critical cells*), and
- pairs $\{\sigma, \tau\}$, where σ is a facet of τ .

A function $f : K \rightarrow \mathbb{R}$ on a cell complex is a *discrete Morse function* if

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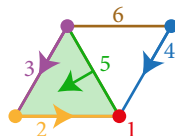
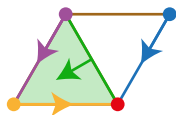
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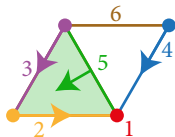
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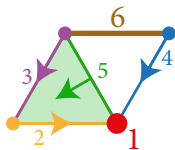
- the sublevel sets $f^{-1}(-\infty, t]$ are subcomplexes, and
- the level sets $f^{-1}(t)$ form a discrete vector field (the *discrete gradient* of f).



Fundamental theorem of discrete Morse theory



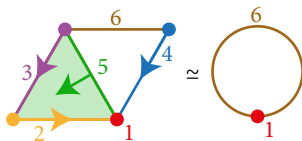
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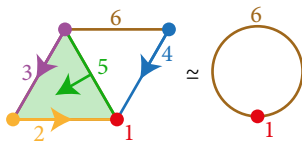
A simplicial complex with a discrete Morse function f is homotopy equivalent to a space (a CW complex) built from the critical simplices of f .



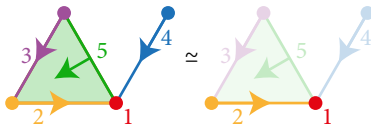
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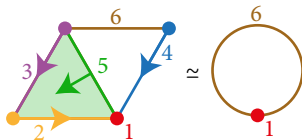
Discrete Morse functions – and their gradients – encode *collapses* of sublevel sets:



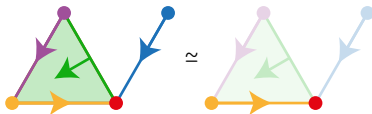
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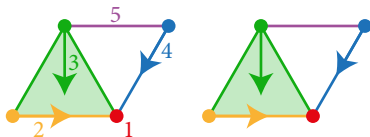


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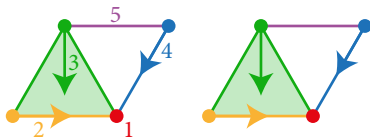
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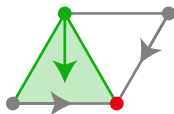


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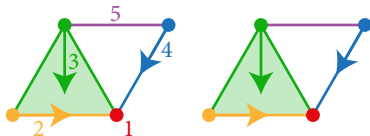


- A generalized vector field V can always be refined to a vector field.

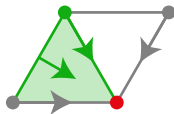


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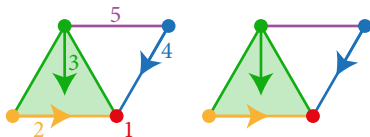


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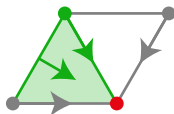


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Proposition (B, Roll 2022)

Let f be a generalized discrete Morse function, breaking ties lexicographically.
Then the apparent pairs of zero persistence form a gradient that

- refines the gradient of f , and
- has the same critical simplices.

The diameter function of generic trees

Proposition (B, Roll 2022)

*Consider a finite weighted tree (V, E) with a generic path length metric (distinct pairwise distances).
Then the diameter function $\text{diam}: \Delta(V) \rightarrow \mathbb{R}$ is a generalized discrete Morse function.*

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In particular, the persistent homology is trivial in degrees > 0 .

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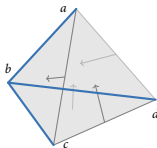
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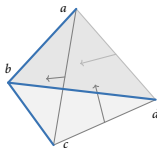
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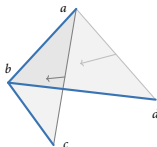
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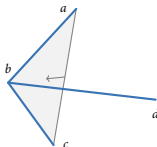
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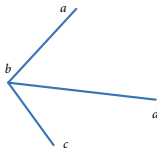
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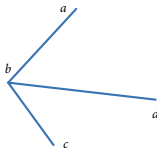
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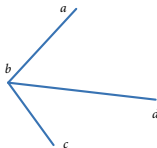
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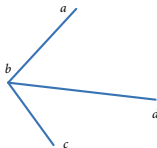
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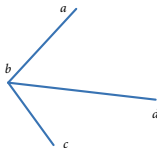
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Wrapping cycles

Morse theory for Čech and Delaunay complexes

Proposition (B, Edelsbrunner 2014)

The Čech complexes and the Delaunay complexes (alpha shapes) are sublevel sets of (generalized) discrete Morse functions. Both functions have the same critical simplices/values.

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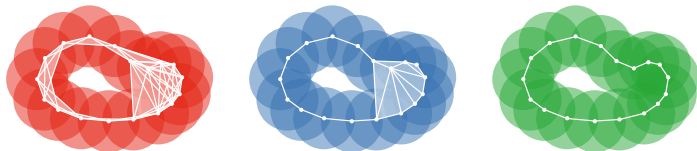
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Theorem (B, Edelsbrunner 2017)

Čech, Delaunay, and Wrap complexes (at any scale r) of a point set $X \subset \mathbb{R}^d$ in general position are related by collapses encoded by a single discrete gradient field:

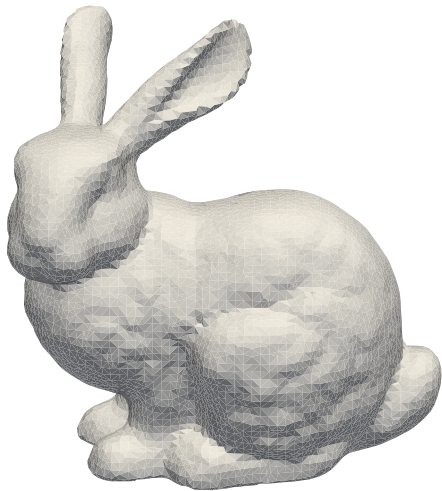
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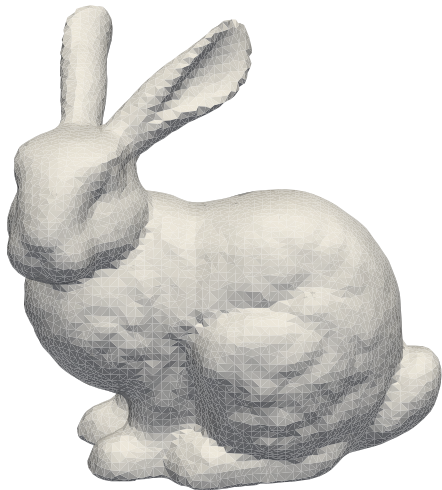
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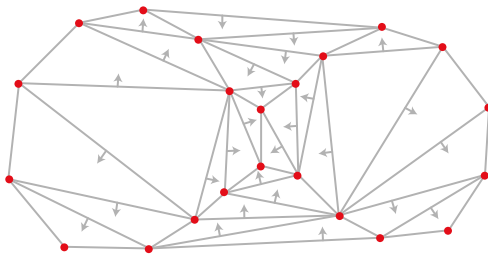


Foundation of the surface reconstruction software *Wrap* (Edelsbrunner 1995, Geomagic)

<http://ulrich-bauer.org/apparent-pairs-talk-lmo.pdf>

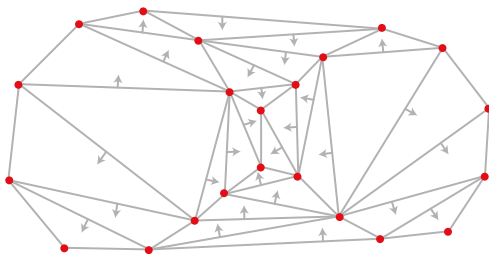
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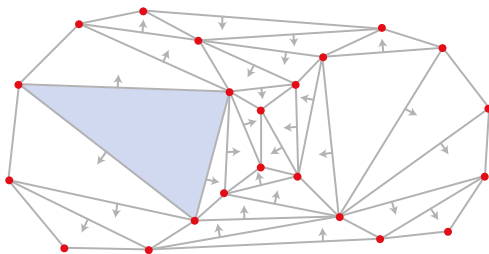
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$\text{Wrap}_r(X)$ is the *descending complex* of V on $\text{Del}_r X$:

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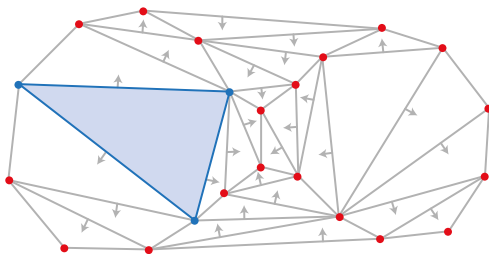
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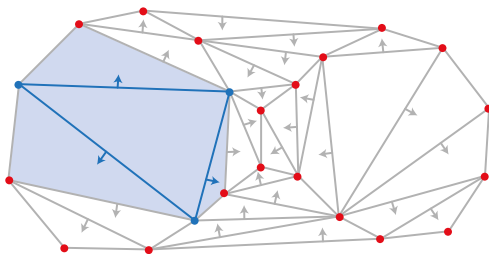
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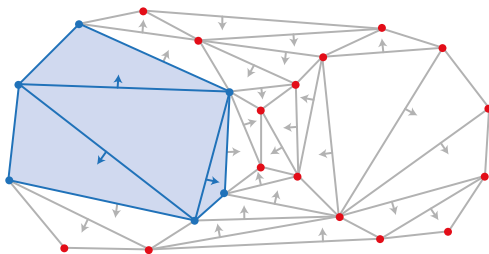
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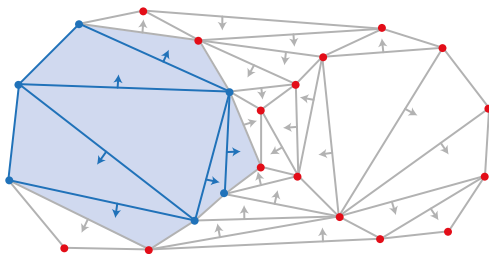
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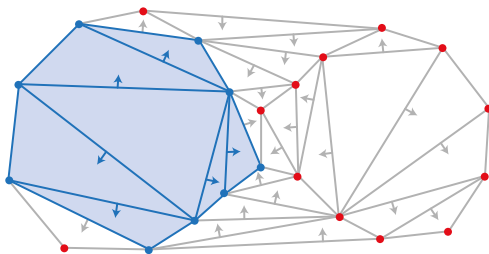
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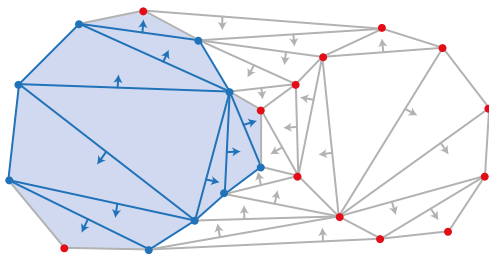
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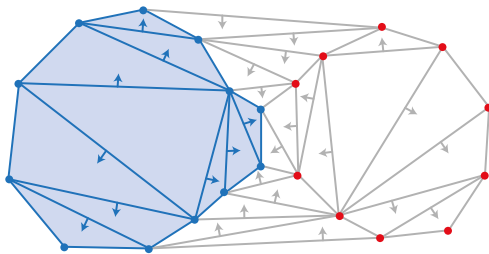
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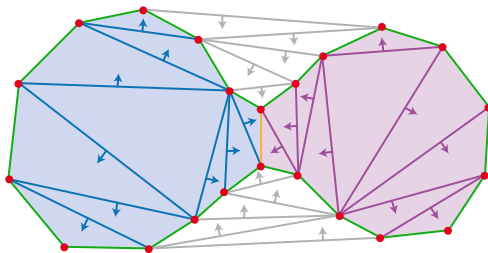
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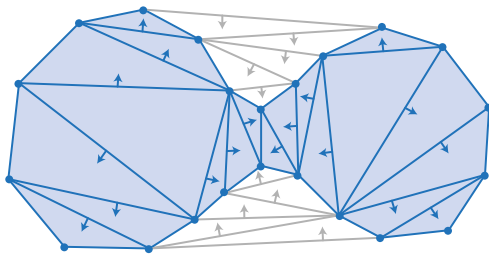
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Algorithm (matrix reduction; a variant of Gauss elimination)

Require: D : $m \times n$ matrix

Ensure: V is full rank upper triangular, $R = D \cdot V$ has unique column pivots

function Reduce(D)

$R = D$

$V = I(n)$

while there exist $i < j$ such that $\text{pivot } R_i = \text{pivot } R_j$ **do**

 add column R_i to column R_j ▷ eliminate the nonzero entry in row $\text{pivot } R_i$

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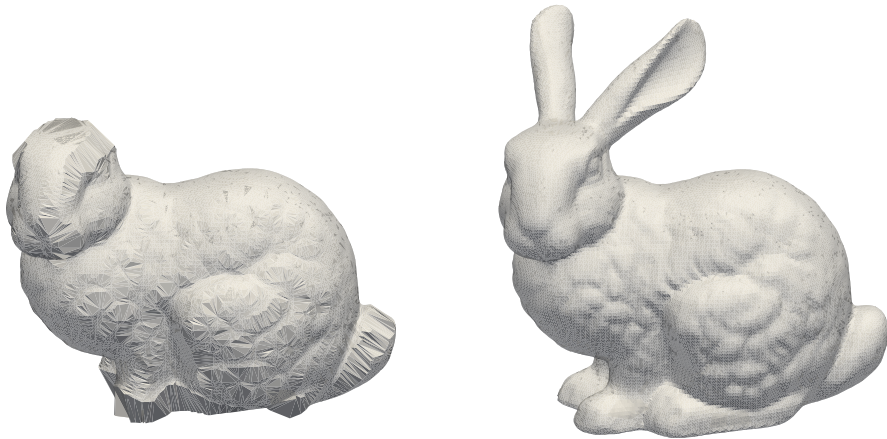
Proposition

The resulting columns R_j are lexicographically minimal cycles within their homology class $[R_j] \in K_{j-1}$.

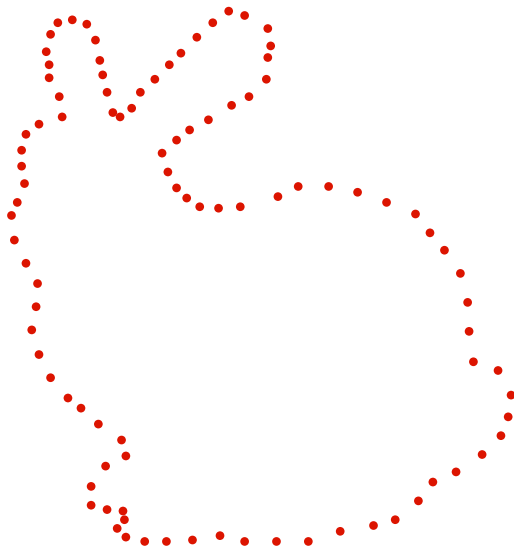
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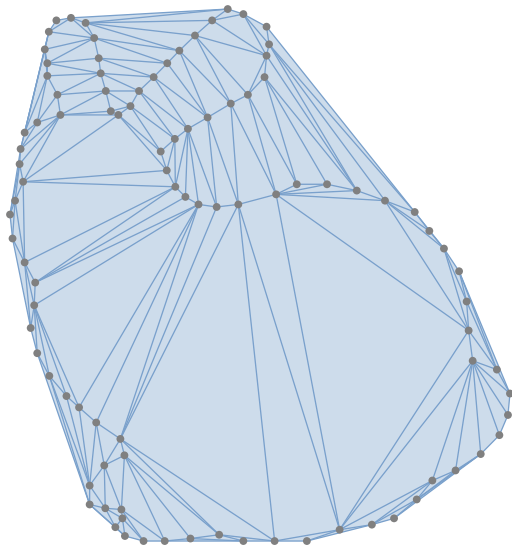
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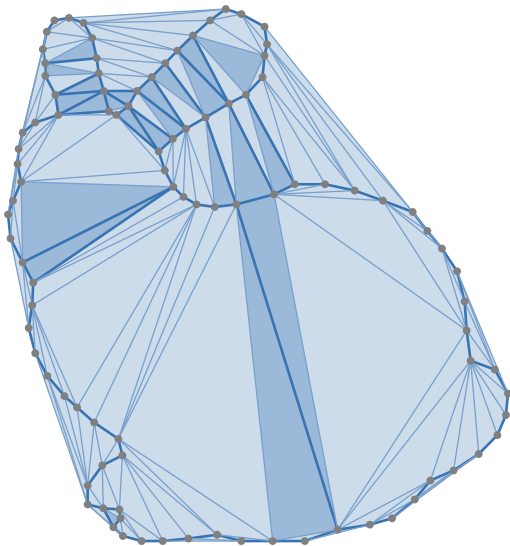
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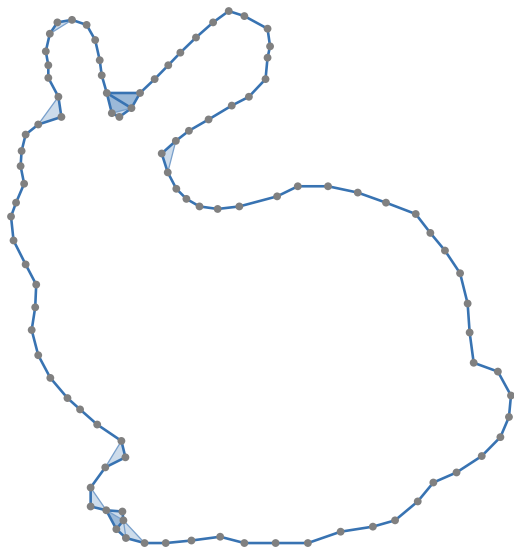
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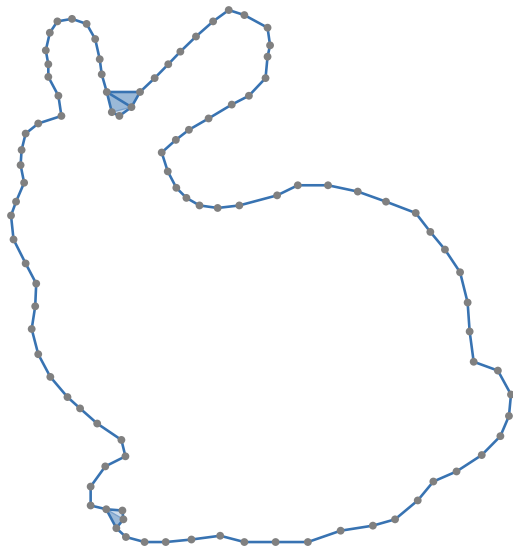
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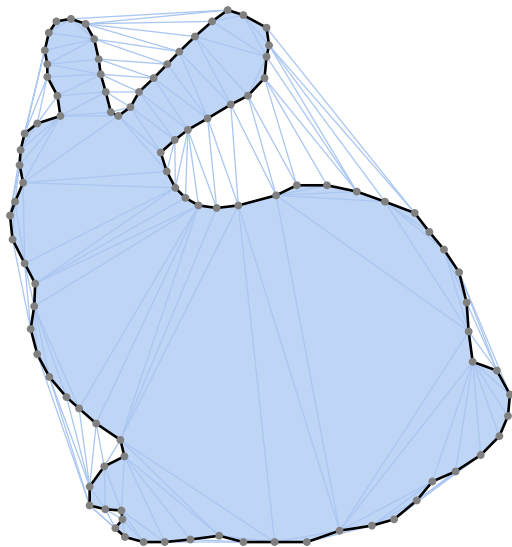
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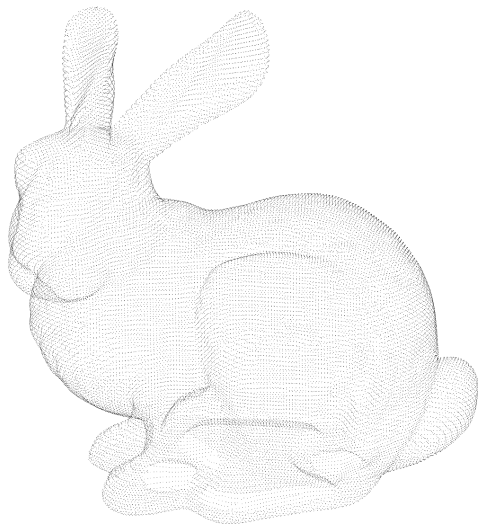
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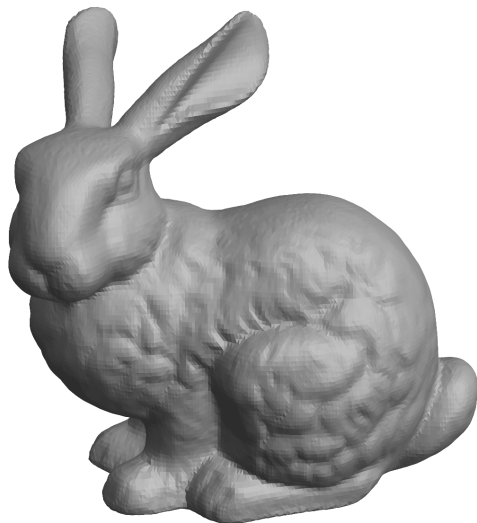
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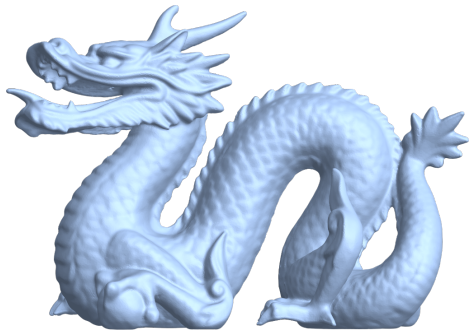
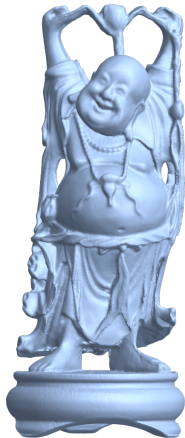
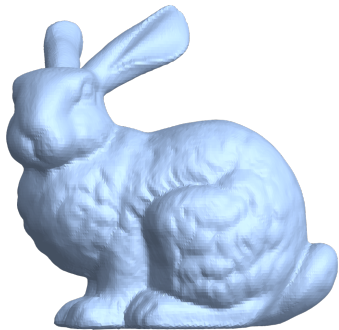
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Surface reconstruction with minimal cycles

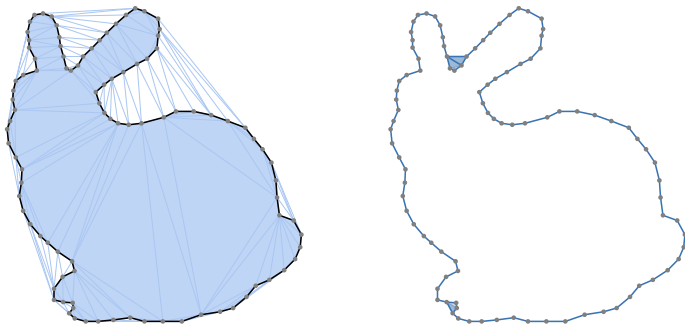


Wrap complexes support minimal cycles

Theorem (B, Roll 2024)

Let $X \subset \mathbb{R}^2$ be a finite subset in general position and let $r \in \mathbb{R}$.

- Exhaustive matrix reduction computes the minimal cycles homologous to a simplex boundary.
- Any lexicographically minimal cycle of $\text{Del}_r(X)$ is supported on $\text{Wrap}_r(X)$.



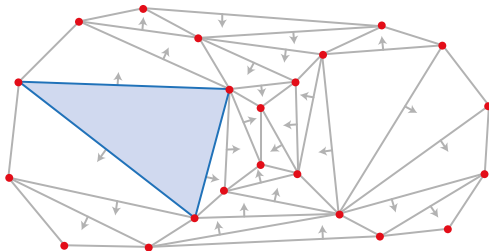
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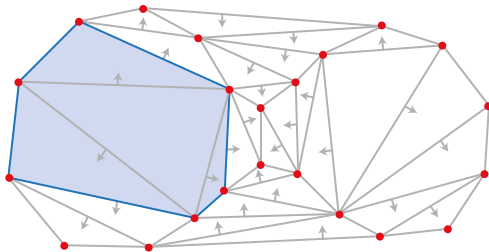
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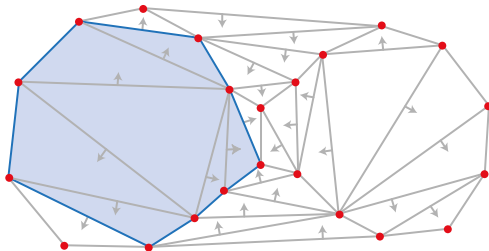
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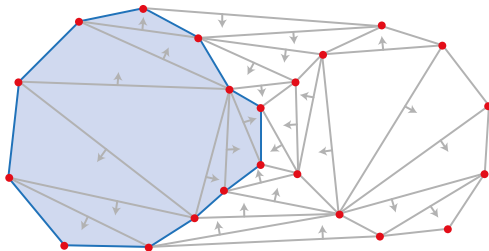
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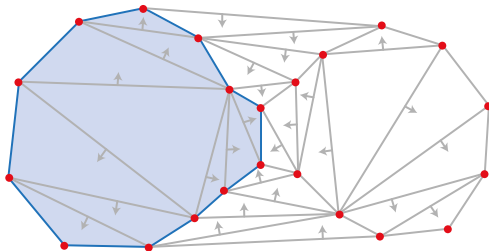
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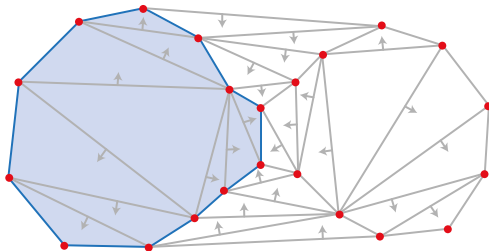


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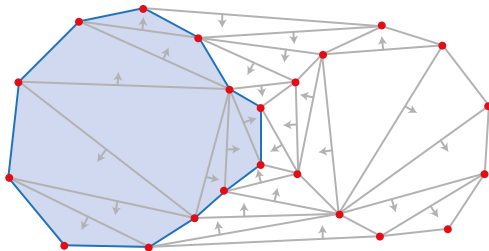
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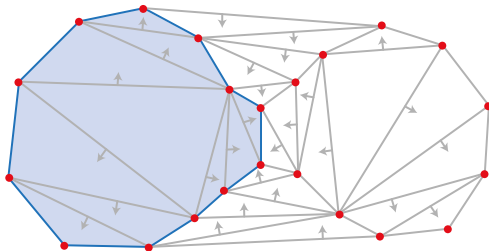
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- Thus the apparent pairs gradient is a subset of the reduction gradient.

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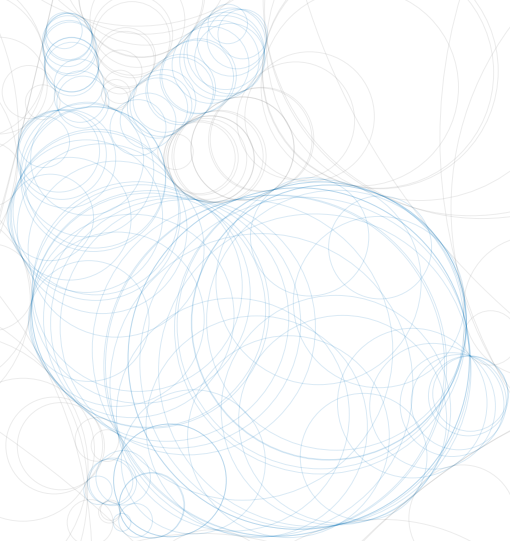
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- Describing the collapse of Rips complexes for trees and hyperbolic spaces
- Illuminating the connection between persistent homology and discrete Morse theory

Thanks for your attention!



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U. Bauer, H. Edelsbrunner

The Morse Theory of Čech and Delaunay Complexes

Transactions of the AMS, 2017. doi:10.1090/tran/6991



U. Bauer

Ripser: efficient computation of Vietoris–Rips persistence barcodes

Journal of Applied and Computational Topology, 2021. doi:10.1007/s41468-021-00071-5



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Gromov hyperbolicity, geodesic defect, and apparent pairs in Vietoris–Rips filtrations

Symposium on Computational Geometry, 2022. doi:10.4230/LIPIcs.SoCG.2022.15



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Wrapping Cycles in Delaunay Complexes: Bridging Persistent Homology and Discrete Morse Theory

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